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Assignment 1

The probability distribution of the amount drawn from an ATM at a Convenience Store is tabulated below, In this table X = the amount withdrawn by a randomly Selected Customer and $F(x) = P[X = x]$. A customer can only withdraw the amounts indicated in the 1st column of the table below.

X	F(x)
10	0.10
20	0.15
40	0.30
60	0.15
80	0.25
100	0.05

a. What is the probability that a randomly Selected Customers withdraws at least \$60 from the ATM?

answer: $F(60) = F(60) + F(80) + F(100)$

$= 0.15 + 0.25 + 0.05$

$= 0.45$ or 45%

b. Show by calculation that the expected value withdrawn from the ATM is \$50 and that the Standard Deviation of the amount withdrawn from this ATM is \$26.27.

$x = 50$?

$SD = 26.27$?

x Qb: Expected value calc incorrect; SD correct

$x = 10, 20, \underbrace{40, 60}, 80, 100$

$\mu = \frac{100}{2}$

$\mu = 50$

Stand deviation =

$E(x^2) = \sum x^2 Px$

$\sigma^2 = 10^2(0.10) + 20^2(0.15) + 40^2(0.30)$

$\sigma^2 = 690$

$\sigma^2 = \sqrt{690} = 26.27$